

Education

B.S in Data Science and Applications

Indian Institute of Technology, Madras

2021-25

Professional Experience

1. Full Stack Engineer | Dom Ventas, Bengaluru

Jan 2026 - Current

- Managing and automating the company's core application that supports operations, retail, marketing, and sales workflows.
- Building an accreditation application integrated with a **RAG-based AI chatbot** for intelligent document retrieval, query handling, and user assistance.
- Maintaining and improving a custom dashboard that connects multiple databases, performs ETL processes, and provides unified business insights.

2. Software Engineer | Artice Inc, CA

Jan 2025 - Jan 2025

- Migrated backend architecture from **Flask to FastAPI**, improving performance, enabling asynchronous processing, and enhancing modularity and maintainability.
- Integrated advanced AI capabilities, including **LLMs, RAG, Text-to-Speech (TTS), and Speech-to-Text (STT)**, to expand intelligent automation features.
- Implemented **WebSockets and webhooks** to enable real-time, bidirectional communication between client and server.

3. AI Engineer | Magizh Technologies, Coimbatore

Jan 2024 - Dec 2024

- Supported backend development of a scalable educational platform for competitive exam preparation, enabling structured content delivery and resource management.
- Developed and integrated an **AI-driven chatbot** providing real-time assistance and personalized learning recommendations to boost engagement.
- Implemented **microservices** for OAuth and multi-channel notifications (Slack, SMS, LinkedIn) to enhance authentication and communication.

4. Web Developer | IIT Madras, Chennai (8 Months)

Jan 2023 - Sept 2023.

- Developed the **"Best Project"** feature on the IITM BS Portal, enabling instructors to add and manage top student projects for each term.
- Built a **Paradox'23 Showcase** page displaying highlights of IIT Madras' annual fest to enhance user engagement.
- Designed responsive, user-friendly pages to ensure seamless access and improved portal experience.

5. Web Developer | IIT Madras, Chennai (8 Months)

Aug 2023 - Oct 2023.

- Collaborated in a team to develop **Vayu Shastra**, a learning app for students enrolled in Vayu Shastra courses.
- Built an Android application using **Flutter** for a smooth, cross-platform user experience.
- Integrated **MongoDB** for backend database management and data storage.

Technical Skills

- Frameworks:** FastAPI, Flask, Django, Vue.js, Bootstrap, Tailwind, WordPress, React (Basics), Next.js, Express.js, Node.js, DRF (Django Rest Framework), Jinja2
- Architecture:** MVC, RBAC, Monolith, Microservices, Event-Driven Architecture, RESTful APIs, GraphQL APIs, Serverless, Multi-tenant Architecture, Clean Architecture
- AI Tools:** n8n, Make.com, Zapier, LangChain, RAG, NLP, SciPy, Hugging Face, LLMs, Tool Calling, MCP, Vector Databases, Prompt Engineering, Fine-tuning, Embeddings, OpenAI APIs, Pinecone/FAISS, AI Agents
- Cloud Technologies:** Azure, GCP, Big Data, Distributed Systems, PySpark, S3 Bucket, AWS (EC2, Lambda, RDS), Kubernetes, Terraform, Cloud Functions, Pub/Sub, Cloud Run
- Databases:** PostgreSQL, MongoDB, ChromaDB, Oracle, MySQL, Elasticsearch, Redis (Caching), DynamoDB, Firebase Firestore, Neo4j (Graph DB), TimescaleDB
- Libraries:** NumPy, Pandas, Streamlit, Selenium (Web Scraping), Matplotlib, Seaborn, Requests, BeautifulSoup, Asyncio, Pydantic
- Programming Languages:** Python, SQL, GraphQL, Java (Basics), JavaScript, HTML5, CSS, TypeScript, Bash/Shell Scripting
- Others:** Kafka, WebSockets, Webhooks, Middlewares, Redis, Celery, Pytest, Unit Testing, RabbitMQ, Nginx, Unicorn, OAuth2, JWT, Authentication/Authorization, Rate Limiting, Caching, Load Balancing
- CI/CD & DevOps:** Docker, Azure, Git, GitHub, GitLab, Postman, CI/CD Pipeline Building, Kubernetes, Helm, GitHub Actions, GitLab CI, Jenkins, Monitoring (Prometheus, Grafana), Logging (ELK Stack)

Personal Projects

Multimodal Clinical AI

- **Developed a clinical documentation pipeline combining Speech-to-Text and Generative AI** for real-time medical transcription.
- **Implemented LLM prompting with RAG and NLP** to generate structured medical reports via FastAPI.

IITM BS Portal integrated with Gen-AI

- **Collaborated with a team to develop and integrate Gen-AI features** into the IITM BS Portal, enhancing user experience through AI-driven insights and tools.
- **Implemented key functionalities** such as lecture and course summarization for last-minute exam prep and personalized question generation based on students' weak topics.

Taxi Fare Guru: Total Amount Prediction Challenge

- **Developed a machine learning model to predict NYC taxi fares** using a dataset of over 55 million records, enhancing accuracy through feature engineering (e.g., distance, time of day, passenger count).
- **Utilized Python with Pandas, Scikit-Learn, and XGBoost** for efficient data preprocessing, model training, and performance evaluation.

Technical Knowledge Base

- **Built** an end-to-end video translation system using OpenAI Whisper (ASR), MarianMT (translation), and Wav2Lip GAN (lip-sync).
- **Containerized** and scaled the pipeline using Docker and Kubernetes.

AI Video Dubbing Pipeline

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- **Utilized Python with Pandas, Scikit-Learn, and XGBoost** for efficient data preprocessing, model training, and performance evaluation.

Unique Mart

- **Designed a grocery management system** with integrated sales forecasting for inventory optimization.
- **Developed data visualization modules** using Flask, SQLite, Matplotlib, and Python.

Course Work

- **Programming & Systems:** Programming in Python, Programming in C, Java Programming, Programming, Data Structures & Algorithms, System Commands (Linux), Database Management Systems, Modern Application Development I & II, Software Engineering, Software Testing
- **Data Science & AI:** Machine Learning Foundations & Techniques, Machine Learning Practice, Deep Learning, Deep Learning Practice, Deep Learning for Computer Vision, Generative AI (Mathematical Foundations), Large Language Models, AI Search Methods, Tools in Data Science, Data Science & AI Lab
- **Mathematics & Statistics:** Mathematics for Data Science I & II, Statistics for Data Science I & II
- **Business & Analytics:** Business Analytics, Business Data Management, Market Research, Design Thinking for Data-Driven App Development
- **Other:** Computational Thinking, Algorithmic Thinking in Bioinformatics, Strategies for Professional Growth

Co-Curricular, Extra-Curricular Activities & Hobbies

- Participated in the **Kharagpur Data Science Hackathon 2024** organized by **IIT Kharagpur**, focusing on data-driven problem solving and model development.
- Competed in **HackCBS 6.0 (2023)**, **India's largest student-run hackathon**, collaborating in a team to build innovative technology solutions under time constraints.
- Passionate about dance and performing arts; performed in dance and drama at the **Paradox** annual fest of **IIT Madras**.
- Enthusiastic about sports and fitness; actively participated in multiple **marathons** and various **sports activities**.